Observability and Audit Evidence Framework Specification

OAEF — System requirements, architecture, deployment and qualification specification, with the MOSIP NIN issuance reference implementation

Field	Value
Document identifier	OAEF-SPEC-001
Version	0.3 — Draft for review
Date	2026-10-19
Work package	WP0 — Shared observability, audit and evidence layer
Owner	[Name, role]
Classification	[To be set by the document owner]
Companion document	OAEF-VTS-001 — Validation Test Specification
Source repository	github.com/gharbonnier78/mosip-observability-oracles

Document control

Version	Date	Author	Change
0.1	2026-10-18	[Author]	First complete draft: generic layer, IaC deployment, qualification layer, MOSIP NIN reference implementation.
0.2	2026-10-18	[Author]	Site attribute: cloud.region (dc1, dc2); DR role oaef.site.role; namespace oaef.* (former open decision D5 resolved).
0.3	2026-10-19	[Author]	Business events (OAEF-SW-12) and continuous profiling (OAEF-SW-13); span links and events in the oracle language; gaps INST-08, INST-09; decisions D7, D8.

Approvals

Role	Name	Date	Signature
Programme / WP0 lead	[ ]	[ ]	
Test and qualification lead	[ ]	[ ]	
Security officer / DPO	[ ]	[ ]	
Platform / operations lead	[ ]	[ ]	

Contents

1 Introduction 4
1.1 Purpose 4
1.2 Scope 4
1.3 Intended audience 4
1.4 Definitions and acronyms 4
1.5 References 5
1.6 Conventions 5
2 System context 6
2.1 Problem statement and goals 6
2.2 Stakeholders and concerns 6
2.3 System context 6
2.4 Operational concept 7
2.5 Constraints 7
2.6 Assumptions and dependencies 7
3 Architecture of the generic layer 8
3.1 Architectural principles 8
3.2 Logical view 8
3.3 Component catalogue 8
3.4 Runtime view 9
3.5 Correlation model 9
3.6 Multi-environment and multi-site view 9
4 Data and interface specification 10
4.1 Signal types and encoding 10
4.2 Attribute registry 10
4.3 Context propagation 11
4.4 Audit event model 11
4.5 Evidence bundle format 11
4.6 Oracle definition language 11
4.7 External interfaces 12
5 Requirements 13
5.1 Software and instrumentation requirements 13
5.2 Qualification layer requirements 14
5.3 Infrastructure requirements 15
5.4 Security and privacy requirements 16
5.5 Telemetry data-quality requirements 16
5.6 Operations and governance requirements 17
5.7 Gap closure 17
6 Deployment and Infrastructure as Code 18
6.1 Deployment topology 18
6.2 Repository layout (GitOps) 18

6.3 IaC toolchain 18
6.4 Configuration baselines 18
6.4.1 Instrumentation (OTel Operator) 18
6.4.2 Collector agent (site stamp) 19
6.4.3 Collector gateway 19
6.4.4 Backends 19
6.5 Environment and site stamping 19
6.6 Promotion and change management 19
6.7 Capacity and retention parameters 20
7 Qualification layer 21
7.1 Concepts 21
7.2 Components 21
7.3 Execution lifecycle 21
7.4 Verdict classification 21
7.5 Reporting and investigation 21
7.6 Golden-run governance 22
8 Reference implementation: MOSIP NIN issuance (NIMS 2.0) 23
8.1 Use case and actors 23
8.2 Activity and rules 23
8.3 Requirement register 23
8.4 Scenario catalogue 24
8.5 Scenario NIN-001 — new adult, no deduplication hit 24
8.5.1 Preconditions and data 24
8.5.2 Interaction sequence 24
8.5.3 Expected signals (extract) 25
8.5.4 Oracle extract 25
8.5.5 Evidence and verdict 25
8.6 Further journeys on the same NIN 25
9 Security and privacy design 26
9.1 Threats considered 26
9.2 Compliance 26
10 Operations 27
11 Verification and traceability 28
12 Open issues and decisions 30
Annex A — Repository index 31

1 Introduction

1.1 Purpose

This document specifies the **Observability and Audit Evidence Framework (OAEF)**: the shared layer that makes every system delivered by the programme produce traces, logs, metrics and audit events according to a common contract, collects them as evidence, and verifies them against exact oracles derived from use cases and business rules. It is the reference for designing, deploying, operating and qualifying that layer, and for every work package that must comply with it.

The framework serves three purposes with one specification:

- **Exact validation** of each delivery against its contractual specification, functional and non-functional, with sealed evidence.
- **Early reports**: realistic expected data allows reports and audit extracts to be designed and tested before the system exists, revealing missing data requirements.
- **Drift detection** over time: data model, business rules, latency, error behaviour, signal completeness and privacy.

1.2 Scope

In scope:

- The **generic layer**: instrumentation conditions, collection pipeline, storage, consumption and correlation model, independent of the business domain.
- Its **deployment as code** (Infrastructure as Code and GitOps) on the development, test/qualification, in-house integration, performance and disaster-recovery platforms, across two sites (DC1, DC2).
- The **qualification layer**: test executor, synthetic data, mocks, evidence collection, oracle engine, verdict classification and reporting.
- A **reference implementation** on MOSIP 1.2.x for the National Identification Number (NIN) issuance journey of NIMS 2.0, including the first scenario NIN-001.

Out of scope: production monitoring service levels, end-user support processes, and the functional design of MOSIP itself.

1.3 Intended audience

Audience	Use of this document
Work-package architects and developers	Instrumentation conditions their components must meet (chapters 4 and 5).
Platform and DevOps engineers	Deployment topology, IaC and configuration baselines (chapter 6).
Test and qualification teams	Qualification layer, oracles and reference scenarios (chapters 7 and 8).
Security officer, DPO, auditors	Privacy, integrity and audit trail design (chapter 9).
Programme management and contracting authority	Requirements, traceability and open decisions (chapters 5, 11 and 12).

1.4 Definitions and acronyms

Term	Definition
Signal	A trace span, log record, metric data point or audit event emitted by a component.
Signal contract	The set of signals a component or a scenario step must emit, with their names, kinds, attributes and links.

Term	Definition
Oracle	A machine-checkable specification of the expected outcome of a scenario: API responses, data states, business rules, signals, audit events and non-functional limits.
Golden run	A reviewed and approved reference execution from which deployment-specific oracle values are frozen.
Evidence bundle	The sealed set of files collected after a scenario run (traces, logs, metrics, audit, business state, context, manifest, verdict).
Verdict class	The automatic classification of a run: PASS, PASS with evidence gap, INCONCLUSIVE, INFRA, SILENT FAILURE, TEST DATA / CONFIG, PRODUCT BUG.
Jarvis	The programme's test executor, which opens the root span of each scenario.
RID / NIN / VID	MOSIP Registration ID; National Identification Number (MOSIP UIN); Virtual ID.
OTel, OTLP	OpenTelemetry; OpenTelemetry Protocol.
IaC, GitOps	Infrastructure as Code; operating model where the desired state in Git is reconciled to the platforms.
WORM	Write once, read many (object lock).
DPIA, NDPA	Data protection impact assessment; Nigeria Data Protection Act 2023.

1.5 References

Ref	Document
[R1]	ISO/IEC/IEEE 29148:2018 — Requirements engineering (structure of this specification).
[R2]	ISO/IEC/IEEE 42010:2022 — Architecture description (views in chapter 3).
[R3]	ISO/IEC/IEEE 29119-3:2021 — Test documentation (companion validation document).
[R4]	OpenTelemetry specification, OTLP 1.x protocol and JSON encoding.
[R5]	OpenTelemetry semantic conventions (resource, HTTP, database, messaging, CICD, test) — version pinned per platform release.
[R6]	W3C Trace Context; W3C Baggage.
[R7]	MOSIP 1.2.x LTS documentation and API specifications (docs.mosip.io; GitHub mosip/*).
[R8]	Nigeria Data Protection Act 2023.
[R9]	OAEF-VTS-001 — Validation Test Specification (companion).
[R10]	Repository mosip-observability-oracles: scenario catalogue, models, oracles, configuration, draw.io diagrams.

1.6 Conventions

- Requirement identifiers: `OAEF-<domain>-<nn>`; domains SW (software and instrumentation), TST (qualification layer), INF (infrastructure), SEC (security and privacy), DQ (telemetry data quality), OPS (operations and governance). They correspond one-to-one to the WP0 identifiers `WP0-<domain>-<nn>`.
- Priority: **M** must, **S** should, **C** could.
- Verification method: **T** test, **D** demonstration, **I** inspection, **A** analysis. Each requirement is verified by at least one test case of [R9].
- Values in **[brackets]** and items marked **[A]** are assumptions or parameters to be agreed; none is a measured value.
- Attributes with no semantic convention use the framework namespace `oaef.*` (decision of 2026-10-18).

2 System context

2.1 Problem statement and goals

In a distributed, event-driven system spanning web and mobile clients, offline registration stations, gateways, many services, message brokers, databases and two data centres, a failed test has three possible causes: a product defect, an infrastructure problem, or missing evidence. Without designed signals these cannot be told apart, and every failure becomes a manual investigation. Retrofitting signals after delivery means changing code and re-running the tests.

Goal	Description
G1	Every delivery emits the signals of its contract, tagged with environment, site, test run and business keys.
G2	Every scenario run produces a sealed evidence bundle and an automatic verdict against an exact oracle.
G3	Every failure is classified to its probable first source and linked to an investigation view.
G4	Personal data never appears in clear in signals; audit trails are immutable and access-controlled.
G5	The same layer serves development, qualification, integration, performance and DR testing on both sites.

2.2 Stakeholders and concerns

Stakeholder	Main concerns
Contracting authority	Acceptance against the contract, objective proof, compliance.
Work-package teams	Clear instrumentation rules, fast feedback, low overhead.
Test and qualification team	Deterministic verdicts, reproducibility, triage time.
Platform operations	Capacity, reliability of the pipeline, maintainability, upgrades.
Security officer, DPO	No PII leakage, access control, integrity, retention limits.
Auditors and regulator	Complete, immutable audit trail linked to business events.

2.3 System context

The framework interfaces with the following external systems (draw.io page 1 shows the full component view).

External system	Interaction	Interface
Systems under test (MOSIP services, stations, portals, apps)	Emit traces, logs, metrics; write audit and business data	OTLP gRPC 4317 / HTTP 4318; MOSIP audit API; databases (read-only access for evidence)
Environment configuration manager (COTS)	Source of environment, site, role and cluster identity	REST API of the product, synchronised to GitOps
CI/CD and test management	Trigger runs, receive results	Pipeline API; result import (JUnit, Cucumber JSON); scenario tags
Partner simulators and partners	ABIS, SMS/email, relying parties, WebSub subscribers	Their protocols; trace context accepted only from authorised partners
Identity provider	Single sign-on and roles for consumers	OIDC
Key management / HSM	Evidence signature	MOSIP Keymanager jwtSign or KMS

2.4 Operational concept

Platform	Purpose	Data	Sampling	Retention (traces / logs / metrics / audit / evidence)
DEV	Developer integration, early contract checks	Synthetic	100 %	[3 d] / [7 d] / [15 d] / [30 d] / —
TEST / QUAL	Qualification campaigns, golden runs	Synthetic	100 % of test traffic	[14 d] / [30 d] / [90 d] / [1 y] / [contract]
INT	In-house integration with simulators and devices	Synthetic	100 %	as TEST
PERF [optional]	Load and soak tests	Synthetic	Tail-based [10 %] + errors + test-tagged	[7 d] / [14 d] / [90 d] / [90 d] / [90 d]
DR-TEST	Multi-site and failover tests on DC1 and DC2	Synthetic	100 %	as TEST

Principal use cases of the framework: **UC-F1** emit signals during a journey; **UC-F2** run a scenario and open its root span; **UC-F3** collect and seal evidence; **UC-F4** verify against the oracle and classify; **UC-F5** investigate from the report; **UC-F6** consult the audit trail; **UC-F7** detect drift across releases.

2.5 Constraints

- MOSIP is a third-party open-source platform: changes are made by agent extension and configuration first; code changes are minimal, documented and proposed upstream.
- OpenTelemetry semantic conventions used here are partly at "development" stability: versions are pinned per platform release.
- Only synthetic personal data may be used on non-production platforms [R8].
- Registration stations operate offline for extended periods [A: up to 72 h].

2.6 Assumptions and dependencies

- [A] Kubernetes is the runtime for MOSIP and for the framework on both sites.
- [A] An environment configuration manager product is available and authoritative for environment identity.
- [A] NIMS 2.0 keeps an 11-digit NIN; the oracle reads the length from configuration.
- Dependency: ABIS vendor capability to propagate trace context through its queues (open decision D4).

3 Architecture of the generic layer

3.1 Architectural principles

ID	Principle	Rationale
P1	Observability first	Signals are designed with the system, because the oracle can only expect what the system can emit.
P2	Standards before custom	OpenTelemetry, W3C Trace Context and semantic conventions wherever they exist; organisation namespace only where none exists.
P3	Extend, do not fork	Agent extensions and configuration rather than changes to MOSIP code.
P4	Where is a resource fact	Environment and site are resource attributes stamped by the cluster collector, not request headers. The site is cloud.region (location); the disaster-recovery role is a separate attribute.
P5	Privacy by design	Identifiers are hashed; PII is denied at source and at the gateway.
P6	Everything as code	Infrastructure, pipeline configuration, dashboards, oracles and scenarios are versioned and reviewed.
P7	Same stores for operations and evidence	Investigation and verification read the same data, so a verdict can always be inspected.

3.2 Logical view

Layer	Responsibility	Main components
L1 Instrumentation	Produce signals with context and business keys	OTel Java agent and MOSIP extension; browser and mobile SDKs; station SDK; gateway tracing; structured logging; audit client
L2 Collection	Receive, enrich, stamp environment and site	OTel Operator; collector agent per node; Prometheus scraping
L3 Pipeline	Sample, redact, derive, route	Collector gateway (tail sampling, redaction, span metrics, service graph, audit routing, load balancing)
L4 Storage	Persist signals, audit and evidence	Tempo; Loki (application and audit tenants); Mimir; object storage with object lock; MOSIP audit database
L5 Consumption	Investigate, alert, audit	Grafana; alerting; auditor access; query helper for hashed identifiers
L6 Qualification	Execute, collect, verify, classify, report	Jarvis; data generator; mocks; evidence collector; oracle engine; reporter

3.3 Component catalogue

Component	Layer	Function	Technology (reference)
Java instrumentation	L1	Spans, metrics, logs for all Java services	OpenTelemetry Java agent, pinned
MOSIP OTel extension	L1	Business keys on spans and MDC; propagation over the event bus	Agent extension jar (to build)
Audit correlation	L1	trace_id/span_id on audit events; audit as OTLP log	Audit client patch
Collector agent	L2	OTLP receiver, Kubernetes metadata, environment/site stamp	OpenTelemetry Collector (DaemonSet)

Component	Layer	Function	Technology (reference)
Collector gateway	L3	Tail sampling, redaction, span metrics, routing	OpenTelemetry Collector (Deployment, HA)
Trace store	L4	Traces, TraceQL	Grafana Tempo
Log store	L4	Logs (app tenant), audit (audit tenant)	Grafana Loki
Metric store	L4	Metrics with exemplars, PromQL	Grafana Mimir
Object storage	L4	Chunks, blocks, evidence bucket, object lock	S3-compatible (MinIO or equivalent)
Grafana	L5	Explore, dashboards, correlations, annotations	Grafana
Jarvis	L6	Scenario execution, root span	Cucumber/TestNG with OTel SDK
Evidence collector	L6	Settle, query, seal, upload	Service or library of Jarvis
Oracle engine	L6	Match oracle, classify verdict	Library with JSON Schemas

3.4 Runtime view

The runtime flow is drawn on draw.io page 6. Summary:

1. Jarvis opens the ROOT span and calls the system under test with W3C `traceparent` and `baggage`.
2. Components emit OTLP spans, logs and metrics, and write audit and business rows.
3. The cluster collector stamps environment and site; the gateway keeps every test trace, redacts PII, derives span metrics and routes audit records to the audit tenant.
4. After a settle window, the evidence collector queries the stores and read-only databases, builds the manifest and seals the bundle in WORM storage.
5. The oracle engine verifies the bundle, classifies the verdict and writes the report with deep links.
6. Investigators open Grafana on the exact trace and time window.

3.5 Correlation model

Key	Links	Survives	Convention
trace_id, span_id	One causal chain	Synchronous calls, propagated messages	W3C Trace Context
cicd.pipeline.run.id, test.case.name, test.suite.name	Test run and scenario	Every span of the journey (baggage)	OTel CICD and Test
Span link (traceId, spanId of the origin)	Asynchronous continuation	Batch jobs, WebSub, ABIS queues	OTel span links
mosip.rid, mosip.uin_hash, mosip.request_id	Business object lifecycle	New traces, batches, audit, database rows	Domain namespace
event.name (business event catalogue)	Business timeline of one RID or NIN	All traces of the journey; durations between steps	OTel Events, domain catalogue
Actor ids (operator, device, user hash)	Human or technical actor	Sessions, channels, offline sync	Domain, hashed
deployment.environment.name, k8s.cluster.name, cloud.region	Where it happened	Everything (resource)	OTel semantic conventions

3.6 Multi-environment and multi-site view

Each site runs its own collection, pipeline and storage stack; evidence and audit buckets replicate from DC1 to DC2. Environment and site identity come from the environment configuration manager, are rendered into GitOps, and are enforced by the collector agent of each cluster (upsert of resource attributes). Cross-site calls appear as parent and child spans with different `cloud.region`, which disaster-recovery oracles assert.

4 Data and interface specification

4.1 Signal types and encoding

Signal	Encoding and transport	Mandatory content
Traces	OTLP over gRPC (4317) or HTTP (4318); JSON encoding: ids lowercase hex, 64-bit integers as strings, enums as integers	traceId, spanId, parentSpanId, name, kind, start/end time, status, resource attributes, required attributes
Logs	OTLP logs (agent logback appender) or JSON stdout parsed by filelog receiver	time, severityNumber/Text, body, traceId, spanId, flags, business keys
Metrics	OTLP metrics; Prometheus scraping for exporters	Standard names and units; exemplars with trace_id
Events	OTLP log records with event.name (OTel Events), emitted inside the producing span	event.name from the catalogue, traceId, spanId, business keys; no personal data (OAEF-SW-12)
Audit	MOSIP audit API to audit database, plus OTLP log to the audit tenant	Audit fields (4.4) plus trace_id, span_id; event name from the catalogue
Profiles	OTel Profiles signal or profiler with span correlation; PERF platform (OAEF-SW-13)	Resource attributes, span_id on samples; informative evidence only

Note: the Tempo JSON API returns trace and span ids in base64; consumers normalise to hex.

4.2 Attribute registry

Attribute	Level	Example	Source
service.name, service.namespace, service.version, service.instance.id	resource	regproc-pkr, mosip, 1.2.0.1	Deployment, operator
deployment.environment.name	resource	test-int	Environment manager
k8s.cluster.name, k8s.namespace.name, k8s.pod.name	resource	dc1-k8s-test	Environment manager, k8sattributes
cloud.region (physical site), cloud.availability_zone (optional)	resource	dc1, dc2; dc1-a	Environment manager. Reuse of the cloud convention for on-premises sites: location only (never a role such as DR), cloud.provider not set, value enforced by the collector stamp.
cicd.pipeline.name, cicd.pipeline.run.id	span (baggage)	run-20261018-001	CI
test.suite.name, test.case.name, test.case.result.status	span	TC-NIN, NIN-001, pass	Jarvis
oaef.test.verdict, oaef.test.persona	ROOT span	PASS, adult-nohit	Jarvis
mosip.rid, mosip.uin_hash, mosip.vid_hash, mosip.request_id, mosip.partner_id, mosip.reg_type, mosip.stage	span, log	29-digit RID; salted hash	MOSIP extension
event.name	event (log)	mosip.registration.packet.received, mosip.identity.activated	Business event catalogue, versioned with the registry

Attribute	Level	Example	Source
HTTP, database, messaging attributes	span	http.response.status_code, db.namespace, messaging.destination.name	Agent (semantic conventions)

4.3 Context propagation

Boundary	Mechanism
HTTP (internal and gateway)	W3C traceparent, tracestate, baggage headers
Kafka / Vert.x event bus	Context in record headers; verified for every regproc stage; extension where the agent does not propagate
JMS (ABIS)	Context in message properties; span link if the vendor does not propagate
Credential batch, WebSub	Originating traceparent stored with the request; span link from the new trace
Offline station packet	traceparent stored in packet metadata, restored by the packet receiver
Partners (egress / ingress)	Baggage stripped on egress; inbound traceparent accepted from authorised partners only

4.4 Audit event model

Field (MOSIP AuditRequestDto / app_audit_log)	Content
eventId, eventName, eventType	Release-specific codes, frozen from the golden run
actionTimeStamp, hostName, hostIp	When and where
applicationId, applicationName, moduleId, moduleName	Which component and stage
sessionUserId, sessionUserName	Actor (pseudonymised where personal)
id, idType	Business reference: RID or NIN hash
trace_id, span_id (added)	Link to the producing span (OAEF-SW-05)

4.5 Evidence bundle format

```
evidence/<run-id>/<scenario-id>/
  manifest.json        ids, time window, environment, site, component versions, exact queries, SHA-256 per
file
  signature.jws        signature of manifest.json
  traces/<trace_id>.json   OTLP JSON
  logs.ndjson          log records (app tenant)
  events.ndjson        business events of the journey, all traces
  audit.json           audit DB rows and audit-tenant records
  profiles/            profiles of spans over budget (optional, informative)
  metrics.json         values at t0 and t_end, exemplars
  business.json        API responses and database states
  context.json         infrastructure and pipeline health in the window
  verdict.json         check results, coverage, verdict class, first source
  links.md             Grafana deep links
```

4.6 Oracle definition language

An oracle is a YAML document with sections `metadata`, `settle`, `rules`, `api`, `db`, `notification`, `traces (with links)`, `logs`, `events`, `metrics`, `audit`, `nfr`, `verdict`. Matchers: literal (typed equality), `$re`, `$cap`, `$ref`, `$in`, `$gte`, `$lte`, `$exists`, `$golden`; span relations `parent`, `parent_ancestor`, `parent_kind`, `after`, `count`,

`link_to`, `link_to_trace`; record-to-span relation `span_of`; collection checks `ordered_subsequence`; event `durations` across traces. Values defined by a standard are explicit; deployment-specific values are frozen from the golden run. The schema of each file type is validated in CI (OAEF-TST-03).

4.7 External interfaces

Interface	Endpoint (reference)	Used by
OTLP ingestion	gRPC :4317, HTTP :4318 /v1/traces, /v1/logs, /v1/metrics	All instrumented components
Trace query	Tempo /api/search (TraceQL), /api/v2/traces/{id}	Evidence collector, Grafana
Log query	Loki /loki/api/v1/query_range, header X-Scope-OrgID	Evidence collector, Grafana
Metric query	Mimir /prometheus/api/v1/query, query_range, query_exemplars	Evidence collector, Grafana
Annotations	Grafana /api/annotations	Oracle engine
Evidence store	S3 API with object lock	Evidence collector, auditors
Environment identity	Environment manager REST API	Sync job, Jarvis

5 Requirements

Each requirement lists its priority, verification method (T test, D demonstration, I inspection, A analysis) and acceptance criterion. Gap identifiers refer to section 5.7.

5.1 Software and instrumentation requirements

ID	Requirement	Pri	Ver.	Acceptance criterion	Closes
OAEF-SW-01	All Java services run the OpenTelemetry Java agent injected by the OTel Operator, one pinned version per platform release, with database semantic-convention opt-in, propagators tracecontext and baggage, OTLP logs and trace-based exemplars.	M	T,I	Every component in the inventory emits spans, metrics and logs with service.name equal to its deployment name; agent version recorded in the evidence manifest.	VER-02
OAEF-SW-02	A MOSIP agent extension sets business keys on spans and in the logging MDC: mosip.rid, mosip.uin_hash, mosip.vid_hash, mosip.request_id, mosip.partner_id, mosip.reg_type, mosip.stage. Never the NIN, VID or other personal data in clear.	M	T	Every regproc, ID repository, IDA, resident and credential span of a test journey carries the expected keys; PII scan finds no clear identifier.	INST-01, INST-04
OAEF-SW-03	Context propagation through the registration processor event bus (Vert.x / Kafka) and JMS (ABIS) is verified; where the agent does not propagate, the extension injects and extracts traceparent and baggage in message headers.	M	T	For NIN-001 every stage consumer span descends from the packet-receiver span in the same trace.	INST-02
OAEF-SW-04	Span links for asynchronous breaks: the credential request stores the originating traceparent and the batch job links to it; WebSub callbacks carry the publisher context and IDA links to it.	S	T	Traces T2 and T3 of NIN-001 each hold a span link resolvable to trace T1.	INST-02
OAEF-SW-05	Audit to trace correlation: the audit client adds trace_id and span_id to each audit event, stored with the event, and each audit event is also emitted as an OTLP log to the audit tenant.	M	T	Every audit row of a test journey is matched to its producing span by id; audit counts in database and audit tenant are equal.	INST-03
OAEF-SW-06	Structured JSON logs on all services: UTC timestamp, level, logger, message key, trace_id, span_id and MDC business keys; stable message keys for oracle matching.	M	T	100 % of log records emitted inside a span carry trace_id and span_id.	INST-01
OAEF-SW-07	Business metrics with OpenTelemetry naming and units: stage processed / failed / reprocessed, UIN and VID pool levels, ABIS round-trip time, credential issuance time, notifications sent and failed, packets queued at stations.	S	T,I	Metrics exist with agreed names and labels; dashboards and alert rules use them.	-
OAEF-SW-08	Registration client instrumented with the OTel Java SDK; the enrolment traceparent is stored with the encrypted packet and restored by the packet receiver; local telemetry buffer bounded and flushed on reconnect; station clock synchronised.	M	T	Offline variant of NIN-001: station and central spans share one trace id after the agreed offline window.	INST-05
OAEF-SW-09	Web and mobile front ends instrumented with OTel browser / mobile SDKs; traceparent on every API call (CORS allows traceparent, tracestate, baggage); consent-aware; no personal data in attributes.	S	T	A resident-portal update journey shows the browser span as root of the backend trace.	INST-06
OAEF-SW-10	The ingress / API gateway creates a server span for every partner call, records mosip.partner_id and accepts inbound	S	T	Bank-authentication scenario shows partner id and remote parent on the first IDA span.	INST-07

ID	Requirement	Pri	Ver.	Acceptance criterion	Closes
	traceparent only from authorised partners; partner simulators are instrumented.				
OAEF-SW-11	Each service exposes build information (version, commit, image digest) via actuator and resource attributes service.version and vcs.ref.head.revision.	M	I	The evidence manifest lists version and digest of every component for each run.	VER-01
OAEF-SW-12	Business events: each business step emits an OpenTelemetry event (log record with event.name and typed attributes) from a versioned event catalogue held in the attribute registry: mosip.registration.packet.received, mosip.registration.stage.completed, mosip.identity.uin.assigned, mosip.identity.activated, mosip.credential.issued, mosip.notification.sent. Events are emitted inside the producing span (trace_id, span_id), carry mosip.rid or mosip.uin_hash and never personal data; audit events follow the same catalogue.	M	T	For NIN-001 the catalogue events are found in order, each joined to its producing span by id; the RID-to-NIN duration is computed from events across traces T1, T2 and T3.	INST-08
OAEF-SW-13	Continuous profiling on the PERF platform (optional on TEST): OpenTelemetry profiles signal or a profiler with span correlation (span_id on samples); when a span exceeds its budget the evidence collector attaches the profile of that service and time window. Profiles never decide a verdict while the signal is not stable.	S	T	A span over its budget in a PERF run links to a CPU profile of the same service and window; the top frames are stored in the evidence bundle.	INST-09

5.2 Qualification layer requirements

ID	Requirement	Pri	Ver.	Acceptance criterion	Closes
OAEF-TST-01	Jarvis opens one ROOT span per scenario with baggage cicd.pipeline.run.id, test.suite.name, test.case.name; step spans as children; results on the ROOT span (test.case.result.status, oaef.test.verdict).	M	T	Every scenario run has exactly one ROOT span with these attributes.	-
OAEF-TST-02	Evidence collector: settle window, queries to traces, logs (app and audit tenants), metrics, read-only databases, business APIs and Kubernetes; manifest with exact queries; SHA-256 per file; signature; upload to the evidence bucket.	M	T	A verdict can be recomputed from the bundle alone (replay).	-
OAEF-TST-03	Oracle engine with matcher language, JSON Schemas for oracle, persona and prerequisite files validated in CI, golden-run record-review-freeze workflow, and classification into seven verdict classes.	M	T	CI rejects an invalid oracle; a golden run needs an approving review; injected faults produce the expected verdict class.	-
OAEF-TST-04	Oracles declare the MOSIP release and agent version they apply to; a platform upgrade triggers re-recording of golden runs and a reviewed diff.	M	T	Running an oracle against a non-matching release stops with ENV_NOT_READY.	VER-01
OAEF-TST-05	Synthetic data generator compliant with the live ID schema and persona contracts (demographics, documents, ISO 19794 biometrics via mock MDS), deterministic by seed, unique against the gallery.	M	T	1 000 generated residents validate against the ID schema; no ABIS hit unless requested.	-
OAEF-TST-06	Instrumented mocks and simulators: ABIS (NO_HIT, HIT, TIMEOUT, FAILURE), SMTP / SMS with read API, relying-party simulator,	M	T	Each mode is selectable through an API and visible in traces.	-

ID	Requirement	Pri	Ver.	Acceptance criterion	Closes
	WebSub subscriber; mode selectable per scenario.				
OAEF-TST-07	Reports with per-clause expected / actual / evidence file, probable first source and Grafana deep link; results exported to the test-management tool by scenario tag.	S	D	Every failed check links to an evidence file and a Grafana URL.	-

5.3 Infrastructure requirements

ID	Requirement	Pri	Ver.	Acceptance criterion	Closes
OAEF-INF-01	Observability stack per site: OTel Operator, collector agent (DaemonSet) and gateway (at least 2 replicas, load balancing by trace id), Tempo, Loki with platform and audit tenants, Mimir, Grafana, Alertmanager.	M	T	Canary signal from every node retrievable in less than [60 s] on each site.	-
OAEF-INF-02	Object storage with object lock (WORM) for audit-tenant data and the evidence bucket, replicated from DC1 to DC2.	M	T	Delete or overwrite on locked objects fails; replica present on DC2.	-
OAEF-INF-03	Tenants per platform class (dev, test, int, perf) plus audit; every signal carries deployment.environment.name, k8s.cluster.name and cloud.region stamped by the cluster collector.	M	T	Environment and site checks ENV-02 and ENV-03 pass on every platform.	-
OAEF-INF-04	Capacity sized for [X] spans/s, [Y] log lines/s and [Z] active series per platform; load-tested before the first campaign.	M	T,A	Collector refused and failed counters stay at 0 at [1.5x] nominal load.	-
OAEF-INF-05	NTP / chrony on all nodes, stations and simulators with offset at most [50 ms].	M	T,I	Offset alert below threshold; oracle clock-skew tolerance never exceeded.	-
OAEF-INF-06	OTLP endpoints internal only, network policies, monitored DC1 to DC2 links, synthetic probes.	M	T	Network policies tested; site-to-site probe metrics available.	-
OAEF-INF-07	Sync job from the environment configuration manager to GitOps (namespace annotations, Instrumentation resource, collector stamp, Jarvis prerequisites) with drift detection.	M	T	Drift check runs before every campaign and on every change, and detects an injected mismatch.	-
OAEF-INF-08	Everything as code (GitOps): collector configuration, dashboards, alert rules, retention, tenants, oracles; changes only by reviewed pull request.	M	T,I	A manual change in the cluster is reverted by the next sync.	-
OAEF-INF-09	Ephemeral test namespaces on demand with pre-seeded masterdata, SoftHSM keys and partners, isolated from shared traffic.	S	T	NIN-001 in an ephemeral namespace passes with exact metric assertions.	ENV-01
OAEF-INF-10	CI runners for Jarvis on each platform with access restricted to MOSIP endpoints, observability APIs and the evidence bucket.	M	T	Runner network policy allows only these destinations.	-
OAEF-INF-11	Backup and disaster recovery of observability stores and evidence.	S	T	Restore test documented; evidence readable from DC2 after loss of DC1.	-

5.4 Security and privacy requirements

ID	Requirement	Pri	Ver.	Acceptance criterion	Closes
OAEF-SEC-01	No personal data in telemetry: deny-list (NIN, VID, names, phone, email, address, biometrics, document numbers) enforced at source and at the gateway; continuous automated PII scan.	M	T	PII scan returns 0 on every run; a seeded PII value is caught (negative test).	-
OAEF-SEC-02	mosip.uin_hash uses the MOSIP salted hash; the investigator lookup helper runs as a restricted service and every lookup is audited.	M	D,I	Lookup audit records exist for each search; the salt is never exposed to browsers or test code.	INST-04
OAEF-SEC-03	SSO (OIDC) and RBAC per platform and tenant; audit tenant readable by auditors only; read-only service accounts for Jarvis and evidence collector; databases through read-only replicas and roles.	M	T	Access matrix tested; denied queries logged.	-
OAEF-SEC-04	TLS (mTLS in the mesh) for OTLP and backend APIs; credentials in a vault; nothing secret in repositories.	M	T	Plain-text connections refused; repository secret scan clean.	-
OAEF-SEC-05	Baggage stripped on egress to third parties; inbound traceparent accepted only from authorised partners; baggage size capped at [8 KB]; no personal data or secrets in baggage.	M	T	Egress capture shows no baggage header to partner simulators.	INST-07
OAEF-SEC-06	Evidence integrity: SHA-256 per file, bundle signature, optional RFC 3161 timestamp, object lock for the retention period.	M	T	Altering one file makes bundle verification fail.	-
OAEF-SEC-07	Synthetic data only on non-production platforms; DPIA of the observability platform; retention limits enforced (NDPA 2023 [A]).	M	I	DPIA approved; retention policies visible in configuration.	-
OAEF-SEC-08	Pinned and signed images for agent, collector and backends; SBOM per release; vulnerability scanning in CI.	S	T	An unsigned image is rejected by the admission policy.	-
OAEF-SEC-09	Artefacts carry their classification; public repositories contain no environment URLs, hostnames, keys or real identifiers.	M	I	Pre-publication scan and manual review recorded for public repositories.	-

5.5 Telemetry data-quality requirements

ID	Requirement	Pri	Ver.	Acceptance criterion	Closes
OAEF-DQ-01	Pinned semantic-convention version and a published attribute registry, including the organisation namespace.	M	T,I	Registry versioned; CI lint rejects unknown custom attributes.	VER-02
OAEF-DQ-02	Signal contract per component checked in CI and on each platform; part of every work package Definition of Done.	M	D	Contract job present in each work package pipeline and blocking.	-
OAEF-DQ-03	Cardinality budget: metric label whitelist (environment, site, cluster, service, route, status); no identifiers as metric labels.	M	A,T	Active series per service under the agreed budget.	-
OAEF-DQ-04	Sampling policy per platform with test-tagged traffic always kept.	M	T	No test-tagged trace missing in the trace store.	-

5.6 Operations and governance requirements

ID	Requirement	Pri	Ver.	Acceptance criterion	Closes
OAEF-OPS-01	Pipeline health objectives (receive / export failures, ingestion lag, canary freshness); a run during a breach is automatically INCONCLUSIVE.	M	T	Injected pipeline failure yields INCONCLUSIVE.	-
OAEF-OPS-02	Provisioned dashboards and alerts: platform health, journey per run, stage RED metrics, audit completeness, PII scan, DR.	S	I	Dashboards and alerts loaded from Git and routed.	-
OAEF-OPS-03	Governance: owners for attribute registry, oracles, golden-run approval and platform; change advisory for agent and convention upgrades.	M	I	Roles and responsibilities published; upgrade checklist used.	-
OAEF-OPS-04	Runbooks and platform support during qualification campaigns, including triage of INCONCLUSIVE and INFRA verdicts.	S	I	Runbooks in Git; support rota during campaigns.	-
OAEF-OPS-05	Onboarding guide and training for work-package teams.	S	D	Each team has executed one scenario end to end.	-

5.7 Gap closure

Gap	Problem	Closed by
INST-01	RID not on spans and logs	OAEF-SW-02, OAEF-SW-06
INST-02	Credential batch and WebSub start new traces	OAEF-SW-03, OAEF-SW-04
INST-03	Audit table has no trace id	OAEF-SW-05
INST-04	No privacy-safe NIN key on IDA, resident and ID repository spans	OAEF-SW-02, OAEF-SEC-02
INST-05	Offline station breaks the trace at sync	OAEF-SW-08
INST-06	Web and mobile front ends do not start or propagate traces	OAEF-SW-09
INST-07	Third-party partner calls are opaque	OAEF-SW-10, OAEF-SEC-05
INST-08	No business-level timeline: log checks rely on free-text bodies and the RID-to-NIN duration spans several traces	OAEF-SW-12, OAEF-SW-04
INST-09	A span over budget shows where time is lost, not which code consumes it	OAEF-SW-13
VER-01	DB columns, stage codes, API paths vary between MOSIP patches	OAEF-SW-11, OAEF-TST-04
VER-02	Span names and attributes vary with agent and convention versions	OAEF-SW-01, OAEF-DQ-01
ENV-01	Shared traffic makes metric deltas inexact	OAEF-INF-09

6 Deployment and Infrastructure as Code

6.1 Deployment topology

Unit	Per	Content
Kubernetes cluster	Platform and site	MOSIP namespaces, framework namespaces (observability, qualification)
Collector agent	Node (DaemonSet)	OTLP receiver, k8sattributes, site stamp
Collector gateway	Cluster (≥ 2 replicas)	Load-balancing exporter tier and sampling tier
Storage stack	Site	Tempo, Loki, Mimir, Grafana, Alertmanager
Object storage	Site, replicated DC1 → DC2	Store buckets; audit and evidence buckets with object lock
CI runners	Platform	Jarvis, evidence collector, oracle engine

6.2 Repository layout (GitOps)

```
platform-gitops/
  terraform/                 clusters, networks, DNS, buckets, object-lock policies
    modules/  envs/dc1/  envs/dc2/
  clusters/<site>-<platform>/    Argo CD applications per cluster
  charts-values/
    otel-operator/  collector-agent/  collector-gateway/
    tempo/  loki/  mimir/  grafana/  alertmanager/
  overlays/<platform>/<site>/     Kustomize: retention, sampling, tenants, env/site stamp (generated)
  dashboards/  alert-rules/  grafana-datasources/
  policies/                  network policies, admission (signed images), RBAC
mosip-observability-oracles/  scenarios, oracles, personas, prerequisites, models (this programme)
```

6.3 IaC toolchain

Concern	Tool (reference)	Rule
Cloud / on-prem resources	Terraform (or OpenTofu)	State remote and locked; plan reviewed in pull request
Cluster workloads	Helm charts + Argo CD	Pinned chart and image versions; automated sync with drift alarm
Per-environment differences	Kustomize overlays	Generated from the environment manager; never edited by hand
Secrets	Vault with external-secrets	No secret in Git
Policy	Admission controller (e.g. Kyverno or Gatekeeper)	Signed images, required labels, resource limits

6.4 Configuration baselines

6.4.1 Instrumentation (OTel Operator)

```
apiVersion: opentelemetry.io/v1alpha1
kind: Instrumentation
spec:
  propagators: [tracecontext, baggage]
  sampler: { type: parentbased_always_on }
  resource:
    resourceAttributes:
      service.namespace: mosip
      deployment.environment.name: test-int      # generated
      k8s.cluster.name: dc1-k8s-test             # generated
```

```
      cloud.region: dc1                          # generated
  java:
    env:
      - { name: OTEL_SEMCONV_STABILITY_OPT_IN, value: database }
      - { name: OTEL_LOGS_EXPORTER, value: otlp }
      - { name: OTEL_METRICS_EXEMPLAR_FILTER, value: trace_based }
```

6.4.2 Collector agent (site stamp)

```
processors:
  k8sattributes: { extract: { otel_annotations: true } }
  resource/site-stamp:
    attributes:
      - { key: deployment.environment.name, value: test-int, action: upsert }
      - { key: k8s.cluster.name,            value: dc1-k8s-test, action: upsert }
      - { key: cloud.region,                value: dc1, action: upsert }
service:
  pipelines:
    traces:  { receivers: [otlp], processors: [memory_limiter, k8sattributes, resource/site-stamp, batch],
exporters: [otlp/gateway] }
```

6.4.3 Collector gateway

```
processors:
  tail_sampling:
    decision_wait: 30s
    policies:
      - { name: keep-test, type: string_attribute, string_attribute: { key: test.case.name, values: [".+"],
enabled_regex_matching: true } }
      - { name: keep-errors, type: status_code, status_code: { status_codes: [ERROR] } }
  redaction/pii: { blocked_values: ["[0-9]{10,11}", "[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+"] }   # illustrative
patterns
  routing/audit: route log records with attribute stream=mosip-audit to the audit exporter
connectors:
  spanmetrics: { resource_metrics_key_attributes: [service.name, deployment.environment.name, cloud.region] }
exporters:
  otlp/tempo, otlphttp/loki-app (X-Scope-OrgID: app), otlphttp/loki-audit (X-Scope-OrgID: audit),
otlphttp/mimir
```

Configuration excerpts are references; exact keys must be checked against the pinned collector and backend versions.

6.4.4 Backends

- Loki: `cloud.region` is a default OTLP index label; add `oaef.site.role`; tenants app and audit; retention per platform.
- Mimir: promotion of `deployment.environment.name`, `k8s.cluster.name`, `cloud.region`, `service.namespace` to series labels.
- Tempo: dedicated columns for environment, site, `mosip.rid`, `test.case.name`.

6.5 Environment and site stamping

A sync job reads each environment record from the environment manager and renders: namespace annotations `resource.opentelemetry.io/<attribute>`, the Instrumentation resource, the collector site stamp and the Jarvis prerequisite file. A drift check compares the rendered values with the live collectors before every campaign (OAEF-INF-07).

6.6 Promotion and change management

- Changes flow DEV → TEST → INT → PERF/DR by pull request and tag; each platform release pins agent, collector, backend and semantic-convention versions.

- An agent or semantic-convention upgrade triggers re-recording and review of golden runs (OAEF-TST-04).

6.7 Capacity and retention parameters

Parameter	Value to agree
Span ingest rate per platform	[X spans/s]
Log ingest rate per platform	[Y lines/s]
Active series per platform	[Z]
Load-test margin	[1.5 ×] nominal
Retention per platform	Section 2.4

7 Qualification layer

7.1 Concepts

A **scenario** is an executable path through a use case, written in Gherkin and bound by tags to a **persona** (data contract), a set of **prerequisites** and an **oracle**. Each run produces an **evidence bundle** and a **verdict**.

7.2 Components

Component	Function
Jarvis executor	Resolves the target environment, checks prerequisites, opens the ROOT span with baggage, executes steps as child spans.
Data generator	Produces residents compliant with the live ID schema and the persona, deterministic by seed, unique against the gallery.
Mocks and simulators	ABIS (NO_HIT, HIT, TIMEOUT, FAILURE), SMTP/SMS with read API, relying party, WebSub subscriber — all instrumented.
Evidence collector	Settle window, queries, manifest, hashing, signature, upload.
Oracle engine	Matches the oracle, computes signal-contract coverage, classifies the verdict.
Reporter	Per-clause results, first source, Grafana links; export to test management.

7.3 Execution lifecycle

1. Resolve environment and site from the environment manager; run prerequisite checks (otherwise ENV_NOT_READY).
2. Record metric baselines; open ROOT span; execute steps.
3. End spans; wait for the settle window (minimum [40 s], two stable polls [15 s] apart, maximum [300 s] otherwise INCONCLUSIVE).
4. Collect traces, logs, metrics, audit, business state and context; build manifest; seal and upload.
5. Verify functional, non-functional and signal-contract sections; classify; report; annotate Grafana.

7.4 Verdict classification

Verdict	Condition	Owner
PASS	Functional checks pass and signal contract 100 %	—
PASS, evidence gap	Functional checks pass, a signal or audit event missing	Observability
INCONCLUSIVE	Evidence pipeline unhealthy or settle window exceeded	Platform, then rerun
INFRA	Functional failure with infrastructure anomaly in the window	Operations
SILENT FAILURE	Functional failure without error signal at the failing step	Development and observability
TEST DATA / CONFIG	Business rejection due to reference data or partner configuration	Test environment
PRODUCT BUG	Error span with exception, infrastructure clean	Development

7.5 Reporting and investigation

The report lists, per contractual clause, the expected value, the actual value and the evidence file; it gives the probable first source and a Grafana deep link scoped to trace, time window, environment and site. When

a span exceeds its budget and a profile is available (OAEF-SW-13), the report adds the top frames of that profile; profiles never change the verdict class. Root-cause analysis then proceeds from the evidence to the relevant code: service code, test code, infrastructure as code, gateway and WAF rules, reference data, or observability configuration.

7.6 Golden-run governance

A golden run is recorded on a validated platform, reviewed by a functional owner and an observability owner, approved by pull request, and frozen into the oracle. It is re-recorded on any platform release that changes agent, semantic conventions or MOSIP version.

8 Reference implementation: MOSIP NIN issuance (NIMS 2.0)

8.1 Use case and actors

Actors: resident applicant, enrolment officer, supervisor, introducer, adjudicator, auditor; enrolment station, central platform (MOSIP), ABIS, messaging gateway, relying party. Use cases: enrol resident, capture biometrics, issue acknowledgement (RID slip), upload packet, validate packet, deduplicate identity, adjudicate possible duplicate, issue NIN and VID, activate identity and issue credential, notify resident, verify NIN, introduce child applicant, consult audit trail. Model: `models/nin-usecase.puml`.

8.2 Activity and rules

The activity (model `models/nin-enrolment-activity.puml`) runs across five lanes: resident, messaging, enrolment station, central platform and ABIS. The station captures data and biometrics, prints the RID slip and uploads the packet, queuing it while offline; the platform validates the packet, deduplicates through the ABIS, routes possible matches to adjudication, otherwise generates the NIN and VID, activates the identity, issues the credential and notifies the resident.

8.3 Requirement register

Tag	Requirement	Verified by
F-01	Demographics and documents validated against ID schema and reference data	NIN-001, NIN-026, NIN-027
F-02	10 fingers, 2 irises, face; quality check; exception path	NIN-001, NIN-002, NIN-029
F-03	Supervisor approval; RID slip issued	NIN-001, NIN-028
F-04	Packet integrity, centre/machine/operator validity, schema conformance; reject with reason	NIN-020 to NIN-027
F-05	Demographic and biometric deduplication via ABIS	NIN-001, NIN-010 to NIN-014
F-06	Possible match to manual adjudication; no NIN until resolved	NIN-010 to NIN-012
F-07	Exactly one NIN (11 digits, check digit [A]) and one perpetual VID	NIN-001
F-08	Identity activated; credential issued to verification partners	NIN-001, NIN-056
F-09	Notification by SMS and email in the chosen language	NIN-001, NIN-004, NIN-074
F-10	Immutable audit event per business step, linked to RID or NIN hash	All
F-11	Relying party verifies NIN or VID	NIN-050 to NIN-055
F-12	Child under five enrolled with introducer	NIN-003
NF-01	Station session ≤ [15] min (adult, no exception) [A]	NIN-001
NF-02	≤ 3 capture retries per modality [A]	NIN-002, NIN-029
NF-03	Offline ≥ [72] h, no data loss, automatic sync [A]	NIN-070, offline NIN-001
NF-04	Upload to NIN ≤ [24] h (no hit, normal load) [A]	NIN-001
NF-05	Notification ≤ [15] min after issuance [A]	NIN-001
NF-06	DR: RPO ≤ [15] min, RTO ≤ [4] h [A]	NIN-080 to NIN-082
NF-07	No PII in clear in signals; audit immutable and access-controlled	All
NF-08	Signal contract 100 %, every signal tagged with environment and site	All, NIN-090 to NIN-093
NF-09	Languages English, Hausa, Yoruba, Igbo [A]	NIN-004

8.4 Scenario catalogue

Family	IDs	Examples
Nominal issuance	NIN-001 to 005	Adult no hit, biometric exception, child, second language, pre-registered
Deduplication	NIN-010 to 014	Demographic hit, biometric hit, adjudication, ABIS failure and timeout
Validation failures	NIN-020 to 029	Upload without sync, hash mismatch, inactive centre, schema violation
Other registration types	NIN-040 to 043	Update, lost NIN, resident update, deactivation
Using the NIN	NIN-050 to 056	Demographic, biometric and OTP authentication; revoked VID; credential
Resilience	NIN-070 to 075	Broker restart, key service down, UIN pool exhausted, SMTP down
Multi-site	NIN-080 to 083	Run on DC2, failover mid-flow, replication, mis-stamped collector
Evidence negatives	NIN-090 to 093	Collector down, audit missing, logs disabled, sampling misconfigured

8.5 Scenario NIN-001 — new adult, no deduplication hit

8.5.1 Preconditions and data

- Prerequisites `env-nin-baseline`: environment resolved from the environment manager; masterdata (centre 10001, machine 10001, operator 110001) active; keys valid; partners and ABIS mock in NO_HIT mode; UIN pool ≥ [1 000]; observability canary retrievable.
- Persona `adult-nohit`: adult 18–80, all demographic fields per ID schema, POA and POI documents, 10 fingers, 2 irises, face (ISO 19794, quality ≥ 80), deterministic seed, unique.

8.5.2 Interaction sequence

Step	From → To	Message (MOSIP 1.2.x)
A1	Jarvis	ROOT span "NIN-001 New adult registration no hit"; baggage cicd.pipeline.run.id, test.case.name
A2	Jarvis → AuthManager	POST /v1/authmanager/authenticate/clientidsecretkey → 200, token
A3	Jarvis → Masterdata	GET /v1/syncdata/latestidschema → ID schema version
B1	Jarvis → Registration status	POST /registrationprocessor/v1/registrationstatus/sync → SUCCESS
B3	Jarvis → Packet receiver	POST /registrationprocessor/v1/packetreceiver/registrationpackets → RECEIVED
B6	Packet receiver → Kafka	MessageDTO {rid, reg_type NEW} with trace context
C1–C8	Stages	Uploader, validator, classifier, CMD/operator validation, demographic and biometric deduplication (ABIS IDENTIFY → 0 candidates)
D1	UIN generator → ID generator	GET /v1/idgenerator/uin → uin
D2	UIN generator → ID repository	POST /idrepository/v1/identity/ → ACTIVATED
D4	→ ID repository	POST /idrepository/v1/vid → perpetual VID
E1–E5	Credential, WebSub, IDA	Credential request, CREDENTIAL_ISSUED event, IDA identity cache
E7	Message sender → Notifier	POST /v1/notifier/email/send, /sms/send
F1–F3	Jarvis	Status search → PROCESSED; identity read; demographic authentication with VID → authStatus true

8.5.3 Expected signals (extract)

Step	Span	Log	Metric	Audit
Upload	SERVER POST …/registrationpackets, mosip.rid; DB insert; Kafka publish	INFO packet received, RID	Request count +1	Packet receiver event
Dedup	JMS producer and consumer, IDENTIFY	INFO 0 candidates	ABIS round-trip	Dedup events
NIN	SERVER GET /v1/idgenerator/uin	INFO UIN assigned (masked)	UIN pool level	UIN generator event
Identity	SERVER POST /idrepository/v1/identity/; encrypt; insert	INFO identity created	Latency, DB pool	Create identity event
Notify	CLIENT POST /v1/notifier/email/send	INFO notification sent	Sent counter	Notification event

8.5.4 Oracle extract

```
- alias: IDGEN_GET_SERVER
  resource: { service.name: idgenerator, cloud.region: ${env.site} }
  name: {$re: "^GET (/v1/idgenerator)?/uin$"}
  kind: 2                       # SERVER
  parent: UINGEN_GET_UIN_CLIENT
  attributes:
    http.response.status_code: 200
    mosip.rid: {$ref: rid}
  duration_ms: {$lte: 500}
```

Full oracle: `oracles/NIN-001.oracle.yaml`; OTLP example: `oracles/examples/NIN-001.otlp-example.json`; feature: `features/nin_001_new_adult_no_hit.feature`.

8.5.5 Evidence and verdict

Expected verdict PASS with signal-contract coverage 100 %, upload-to-PROCESSED within the agreed limit, no ERROR span or log, no PII in clear, audit multiset identical in database and audit tenant.

8.6 Further journeys on the same NIN

Filtering on `mosip.uin_hash` returns every journey of one identity: issuance (NIN-001), third-party authentication by a relying party (NIN-050/051, IDA with Keymanager and identity cache) and resident self-service update (NIN-042, resident service, OTP validation, update packet, ID repository update). Each is a separate trace; the journey view is assembled by query.

9 Security and privacy design

9.1 Threats considered

Threat	Control
PII leakage through spans, logs or metrics	Deny-list at source and gateway; continuous PII scan; hashing of identifiers (OAEF-SEC-01/02)
Re-identification through NIN search	Restricted hashing helper; every lookup audited (OAEF-SEC-02)
Unauthorised access to audit trail	SSO and RBAC; audit tenant restricted to auditors (OAEF-SEC-03)
Tampering with evidence or audit	Hashes, signature, object lock (OAEF-SEC-06)
Context injection from outside	Inbound context only from authorised partners; baggage size limit (OAEF-SEC-05)
Compromised images or agents	Signed images, SBOM, admission policy (OAEF-SEC-08)
Disclosure through public repositories	Classification and pre-publication scan (OAEF-SEC-09)

9.2 Compliance

Only synthetic data is used outside production; a DPIA covers the observability platform; retention limits are enforced by configuration [R8].

10 Operations

- Pipeline health objectives: zero refused or failed exports at nominal load, ingestion lag below [30 s], canary freshness below [60 s]; any breach marks runs INCONCLUSIVE.
- Provisioned dashboards: platform health, journey per run, stage RED metrics, audit completeness, PII scan, DR.
- Governance: owners for the attribute registry, oracles, golden-run approval and the platform; change advisory for agent and convention upgrades.
- Runbooks and on-call support during qualification campaigns; onboarding of work-package teams.

11 Verification and traceability

Every requirement of chapter 5 is verified by test cases of the Validation Test Specification [R9]. The matrix below gives the primary test case per requirement.

Requirement	Pri	Ver.	Validation test cases (OAEF-VTS-001)
OAEF-SW-01	M	T,I	VT-SW-01, VT-E2E-01
OAEF-SW-02	M	T	VT-SW-02, VT-E2E-01
OAEF-SW-03	M	T	VT-SW-03, VT-E2E-01
OAEF-SW-04	S	T	VT-SW-04, VT-SW-10
OAEF-SW-05	M	T	VT-PIPE-04, VT-SW-05, VT-SW-10, VT-E2E-01, VT-E2E-02
OAEF-SW-06	M	T	VT-SW-02, VT-E2E-01
OAEF-SW-07	S	T,I	VT-PIPE-05, VT-SW-06
OAEF-SW-08	M	T	VT-SW-07
OAEF-SW-09	S	T	VT-SW-08
OAEF-SW-10	S	T	VT-SW-09
OAEF-SW-11	M	I	VT-PIPE-01, VT-SW-01
OAEF-SW-12	M	T	VT-PIPE-07, VT-SW-10, VT-E2E-01
OAEF-SW-13	S	T	VT-SW-11
OAEF-TST-01	M	T	VT-QUAL-01, VT-E2E-01
OAEF-TST-02	M	T	VT-PIPE-01, VT-SEC-06, VT-QUAL-02, VT-E2E-01
OAEF-TST-03	M	T	VT-PIPE-06, VT-QUAL-03, VT-QUAL-08, VT-E2E-01
OAEF-TST-04	M	T	VT-QUAL-04
OAEF-TST-05	M	T	VT-QUAL-05
OAEF-TST-06	M	T	VT-QUAL-06
OAEF-TST-07	S	D	VT-QUAL-07
OAEF-INF-01	M	T	VT-INF-01
OAEF-INF-02	M	T	VT-INF-02
OAEF-INF-03	M	T	VT-INF-03, VT-E2E-01, VT-E2E-03
OAEF-INF-04	M	T,A	VT-INF-04
OAEF-INF-05	M	T,I	VT-INF-05, VT-SW-07
OAEF-INF-06	M	T	VT-INF-06
OAEF-INF-07	M	T	VT-INF-03, VT-INF-07
OAEF-INF-08	M	T,I	VT-INF-08
OAEF-INF-09	S	T	VT-INF-09
OAEF-INF-10	M	T	VT-INF-06
OAEF-INF-11	S	T	VT-INF-10, VT-E2E-03
OAEF-SEC-01	M	T	VT-PIPE-03, VT-SEC-01, VT-E2E-01
OAEF-SEC-02	M	D,I	VT-SEC-02
OAEF-SEC-03	M	T	VT-PIPE-04, VT-SEC-03
OAEF-SEC-04	M	T	VT-SEC-04
OAEF-SEC-05	M	T	VT-SW-09, VT-SEC-05
OAEF-SEC-06	M	T	VT-INF-02, VT-SEC-06

Requirement	Pri	Ver.	Validation test cases (OAEF-VTS-001)
OAEF-SEC-07	M	I	VT-SEC-07
OAEF-SEC-08	S	T	VT-SEC-08
OAEF-SEC-09	M	I	VT-SEC-09
OAEF-DQ-01	M	T,I	VT-PIPE-07, VT-DQ-01
OAEF-DQ-02	M	D	VT-DQ-02, VT-E2E-02
OAEF-DQ-03	M	A,T	VT-PIPE-05, VT-DQ-03
OAEF-DQ-04	M	T	VT-INF-04, VT-PIPE-01, VT-PIPE-02
OAEF-OPS-01	M	T	VT-INF-01, VT-PIPE-06, VT-QUAL-08
OAEF-OPS-02	S	I	VT-OPS-01
OAEF-OPS-03	M	I	VT-OPS-02
OAEF-OPS-04	S	I	VT-OPS-02
OAEF-OPS-05	S	D	VT-OPS-02

12 Open issues and decisions

Resolved on 2026-10-18: the site attribute is `cloud.region` (`dc1`, `dc2`); the disaster-recovery role is `oaef.site.role`; attributes with no semantic convention use the framework namespace `oaef.*`.

ID	Decision required
D1	Environment configuration manager product and record ownership.
D2	Tenant model: shared stack with tenants per platform class (proposed) or one stack per platform.
D3	Handling of MOSIP code changes not achievable by extension (audit trace id, credential traceparent, packet traceparent).
D4	ABIS vendor propagation of trace context, or span link boundary.
D5	Retention, capacity and threshold values (all bracketed values).
D6	Publication status of the specification repository.
D7	Owner of the business event catalogue; audit events emitted as catalogue events by the audit client, or mapped by the collector from the audit stream.
D8	Profiling tool (OTel Profiles with eBPF profiler, or async-profiler to Pyroscope) and whether a performance verdict class is added once the Profiles signal is stable.

Annex A — Repository index

Path	Content
models/	Use case and activity (PlantUML); requirement register
catalog/TC-NIN-scenarios.md	Scenario catalogue
features/	Gherkin scenarios
oracles/	Oracles and OTLP examples
data/personas/	Persona data contracts
prerequisites/	Environment prerequisites
observability/	Attribute model, injection and backend configuration
docs/diagrams/	draw.io: architecture, E2E flow, NIN sequence, expected signals, evidence and verdict, observability and audit flow
docs/wp0-shared-platform-requirements.md	WP0 requirements (source of chapter 5)